

Resham J. Joshi

MSc Computer Science (pursuing)

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Fraunhofer UMSICHT Institute

Sulzbach-Rosenberg, Germany.

Unsolicited application Requisition Number 83413

Dear Ms. Herbst, dear Ms. Wagner,

I'm applying on my own initiative because I want to work at Fraunhofer again. Fraunhofer IGCV in Augsburg was my first job in Germany, and it shaped my interest in applied research. I'd like to bring that experience to Sulzbach-Rosenberg.

At IGCV I spent close to two years taking problems from a first question to a working system clarifying requirements, comparing methods, and building complete machine learning pipelines. The work covered anomaly detection, causal analysis, graph neural networks, dashboards, REST APIs, and Docker based deployment. My reference letter from IGCV is attached.

I'm currently in software development at the Technical University of Munich. Alongside my regular tasks, I proposed a set of portal improvements which includes a chatbot to guide users through complex processes on portal which was accepted and taken forward as a project in our team.

I like looking past the obvious solution, spotting improvements, and turning ideas into something that actually runs. Let me be honest about fit I'm not sure how much and how deep I know about the topics you have been working on, and I don't yet know every research topic at UMSICHT, but I am ready to learn new things and also I like to take challenges, so I might be good fit according to me.

As far as I understood your work, it runs on plant and sensor data pilot pyrolysis and TCR plants, combustion and process heat, LCA and secondary resource assessment and that's exactly where my kind of work is useful like detecting faults and drift early, supporting root cause analysis, predicting process outcomes, and making research data easier to query.

One example where I can help might be with the scattered knowledge graphs making them structured, so information can be retrieved effectively and faster and they can be used across different projects. My master's thesis also is in the area of improving quality of Knowledge graphs without domain experts using Reinforcement learning.

These are starting points but I want to understand your teams' real needs first, and I am ready to learn the domain knowledge that takes. I recently completed my Thesis for my MSc in Computer Science at RPTU Kaiserslautern.

My German is currently B1, and I keep working on it:

Ich kann sehr gut auf Deutsch gut verstehen und ich muss üben. Ich möchte mein Deutsch verbessern.

I would like to have a conversation to know about the tasks your teams are working on now and where I could contribute.

Kind regards,

Resham Joshi